习题10.1:1,2,3(1)(2),4(1)(2),5,6(2)(3).
10.2:1,2,3,4,5,7,9,10
10.3:2(3)(4),3,4,5(1)(5),6,9,11.

10.1

10.1.1

Suppose f is bounded on $[a,b]$. If for any $a' \in (a,b)$, $f \in R[a',b]$, prove that $f \in R[a,b]$ and

$$\lim_{a' \to a^+} \int_{a'}^{b} f \,\mathrm{d}x = \int_{a}^{b} f \,\mathrm{d}x.$$

Proof: Suppose $|f| \leqslant M$ on $[a,b]$. Then

$$-M(a'-a) + \underline{\int_{a'}^{b}} f dx \leqslant \underline{\int_{a}^{b}} f dx \leqslant \overline{\int_{a}^{b}} f dx \leqslant M(a'-a) + \overline{\int_{a'}^{b}} f dx$$

Hence

$$\overline{\int_{a}^{b}} f dx - \underline{\int_{a}^{b}} f dx \leqslant 2M(a'-a) \to 0,$$

so $f \in R[a,b]$, and $\lim_{a' \to a} \int_{a'}^{b} f \,\mathrm{d}x = \int_{a}^{b} f \,\mathrm{d}x$.

10.1.2

Prove that

$$\frac{\pi}{2\sqrt{2}} < \int_0^1 \frac{\mathrm{d}x}{\sqrt{1-x^4}} < \frac{\pi}{2}.$$

Proof: Note that

$$\frac{\pi}{2\sqrt{2}} < \int_0^1 \frac{\mathrm{d}x}{\sqrt{2} \cdot \sqrt{1-x^2}} < \int_0^1 \frac{\mathrm{d}x}{\sqrt{1-x^4}} < \int_0^1 \frac{\mathrm{d}x}{\sqrt{1-x^2}} = \frac{\pi}{2}.$$

10.1.3

Calculate
(1)

$$\int_2^{\infty} \frac{\mathrm{d}x}{x(\log x)^2} = \int_{\log 2}^{\infty} \frac{\mathrm{d}y}{y^2} = \frac{1}{\log 2}.$$

(2)

$$\int_1^\infty \frac{\log x}{x^2}\,\mathrm{d}x = \int_0^\infty ye^{-y}\,\mathrm{d}y = \int_0^\infty e^{-y}\,\mathrm{d}y = 1.$$

10.1.4

Calculate
(1)

$$\int_0^1 \frac{x^3}{\sqrt{1-x^2}}\,\mathrm{d}x = \int_0^{\pi/2} \sin^3\theta\,\mathrm{d}\theta = -\int_0^{\pi/2} (1-\cos^2\theta)\,\mathrm{d}\cos\theta = \int_0^1 1-t^2\,\mathrm{d}t = \frac{2}{3}.$$

(2)

$$\int_{-1}^1 \frac{x}{\sqrt{1-x^2}}\,\mathrm{d}x = 0.$$

10.1.5

Determine convergence:
(1)

$$\int_0^\infty |\cos x^2|\,\mathrm{d}x = \int_0^\infty \frac{|\cos y|}{2\sqrt{y}}\,\mathrm{d}y = \infty.$$

(2)

$$\int_0^\infty \frac{\mathrm{d}x}{e^x\sqrt{x}} < \int_0^1 \frac{\mathrm{d}x}{\sqrt{x}} + \int_1^\infty \frac{\mathrm{d}x}{x^{3/2}} < \infty.$$

10.1.6

Determine convergence:
(2)

$$\int_0^1 \frac{\log x}{1-x}\,\mathrm{d}x = \int_0^1 \frac{\log(1-x)}{x}\,\mathrm{d}x = -\frac{\pi^2}{6}.$$

(3)

$$\int_0^2 \frac{\mathrm{d}x}{(x-1)^2} = 2\int_0^1 \frac{\mathrm{d}x}{x^2} = \infty.$$

10.2

10.2.1

Determine convergence:
(1)

$$\int_1^\infty \frac{\mathrm{d}x}{(1+x^4)^{1/3}} \leqslant 4^{4/15} \int_1^\infty \frac{\mathrm{d}x}{x^{16/15}} < \infty.$$

(2)

$$\int_0^\infty \frac{\mathrm{d}x}{1+\sqrt{x}} > \frac{1}{2} \int_1^\infty \frac{\mathrm{d}x}{\sqrt{x}} = \infty.$$

(3)

$$\int_0^\infty \frac{x^3}{\sqrt{1+x^7}}\,\mathrm{d}x > \frac{1}{2} \int_1^\infty x^{-1/2}\,\mathrm{d}x = \infty.$$

10.2.2

Determine convergence:

(1)

$$\int_0^1 \frac{\mathrm{d}x}{(1-x^3)^{1/3}} < \int_0^1 \frac{\mathrm{d}x}{(1-x)^{1/3}} = \frac{3}{2}.$$

(2)

$$-\int_0^1 \frac{\mathrm{d}x}{\sqrt{x}\log x} = \int_0^\infty \frac{e^{-x/2}}{x}\,\mathrm{d}x > \int_0^1 \frac{e^{-1/2}}{x}\,\mathrm{d}x = \infty.$$

(3)

$$\int_0^1 \frac{\mathrm{d}x}{\sqrt{1-x^4}} < \int_0^1 \frac{\mathrm{d}x}{\sqrt{1-x^2}} = \frac{\pi}{2}.$$

10.2.3

Determine convergence and absolute convergence.

(1)

$$\int_1^\infty \frac{\sin\sqrt{x}}{x}\,\mathrm{d}x = 2\int_1^\infty \frac{\sin y}{y}\,\mathrm{d}x$$

converges but not absolutely.

(2)

$$\int_1^\infty \frac{\cos x}{x}\,\mathrm{d}x = \int_1^\infty \frac{\mathrm{d}\sin x}{x} = \sin 1 + \int_1^\infty \frac{\sin x}{x^2}\,\mathrm{d}x$$

converges, but not absolutely.

(3)

$$\int_e^\infty \frac{\log\log x}{\log x}\sin x\,\mathrm{d}x = \int_1^\infty \frac{\log t}{t}\sin e^t\,\mathrm{d}e^t = \int_1^\infty \frac{\log t}{t}\,\mathrm{d}\cos e^t = \int_1^\infty \cos e^t \frac{\log t - 1/t^2}{t^2}\,\mathrm{d}x < \infty.$$

converges.

10.2.4

Determine convergence:
(1)

$$\int_0^\infty \sin x^p \,\mathrm{d}x = \frac{1}{p}\int_0^\infty \frac{\sin y}{y^{1-1/p}}\,\mathrm{d}y$$

converges when $p > -1$ and $1 - 1/p > 0$, i.e. $p > 1$.
(2)

$$\int_0^\infty \frac{\cos x}{x^p}\,\mathrm{d}x$$

converges when $0 < p < 1$.
(3)

$$\int_0^\infty \frac{1-\cos x}{x} e^{-x}\,\mathrm{d}x < \int_0^1 x e^{-x}\,\mathrm{d}x + \int_1^\infty 2e^{-x}\,\mathrm{d}x.$$

converges.

10.2.5

Suppose $f \in C[1,\infty)$. If $\int_1^\infty f^2\,\mathrm{d}x$ converges, prove that $\int_1^\infty \frac{f(x)}{x}\,\mathrm{d}x$ converges.
Proof:

$$\int_1^\infty |\frac{f(x)}{x}|\,\mathrm{d}x \leqslant \frac{1}{2}\int_1^\infty f^2 + x^{-2}\,\mathrm{d}x < \infty.$$

10.2.6

Suppose $f \in R((0,b))$ for all $b > 0$, and $\lim_{x\to\infty} f(x) = 0$, prove that

$$\lim_{b\to\infty}\frac{1}{b}\int_0^b f\,\mathrm{d}x = 0.$$

Proof: Suppose $\varepsilon(N) = \sup_{x>N}|f(x)|$, then $\lim_{N\to\infty}\varepsilon(N) = 0$. Let $F(b) = \frac{1}{b}\int_0^b f\,\mathrm{d}x$, then $|F(b+T)| \leqslant \frac{b}{T}|F(b)| + \varepsilon(b)$. Hence $\limsup_{x\to\infty}|F(x)| \leqslant \varepsilon(b)$ for any $b > 0$, so $F(x) \to 0$ and $\lim_{b\to\infty}\frac{1}{b}\int_0^b f\,\mathrm{d}x = 0$.

10.2.7

Suppose $f \in C[0,\infty)$. If $f > 0$ and $\int_0^\infty \frac{\mathrm{d}x}{f}$ converges, prove that

$$\lim_{b\to\infty}\frac{1}{b^2}\int_0^b f\,\mathrm{d}x = \infty.$$

Proof: Note that

$$\int_b^{2b} f\,\mathrm{d}x \cdot \int_b^{2b} \frac{1}{f}\,\mathrm{d}x \geqslant b^2,$$

so

$$\frac{1}{b^2}\int_0^b f\,\mathrm{d}x > \frac{1}{b^2}\frac{b^2}{4} \Big/ \int_{\frac{2}{b}}^{\infty} \frac{1}{f}\,\mathrm{d}x \to \infty.$$

10.2.9

Suppose γ is the Euler constant, prove that

$$\int_1^{\infty} \frac{1}{[x]} - \frac{1}{x}\,\mathrm{d}x = \gamma.$$

Proof:

$$\int_1^{\infty} \frac{1}{[x]} - \frac{1}{x}\,\mathrm{d}x = \sum_{n=1}^{\infty}\int_n^{n+1} \frac{1}{n} - \frac{1}{x}\,\mathrm{d}x = \sum_{n=1}^{\infty}\frac{1}{n} - \log\frac{n+1}{n} = \lim_{N\to\infty}\sum_{n=1}^{N}\frac{1}{n} - \log N = \gamma.$$

10.2.10

Determine convergence of

$$\int_0^{\infty} \frac{\mathrm{d}x}{1+x^4\sin^2 x}$$

Solution:
Note that

$$\int_0^{\infty} \frac{\mathrm{d}x}{1+x^4\sin^2 x} < C + \sum_{n=1}^{\infty}\int_{n\pi}^{(n+1)\pi} \frac{\mathrm{d}x}{1+(n\pi)^4\sin^2 x} < C + \frac{C'}{\pi^2}\sum_{n=1}^{\infty}\frac{1}{n^2} < \infty.$$

where

$$\int_0^{\pi/2} \frac{\mathrm{d}x}{1+a^2\sin^2 x} < \int_0^{\pi/2} \frac{\mathrm{d}x}{1+(2a/\pi)^2x^2} < \frac{\pi}{2a}\cdot\frac{\pi}{2}.$$

Hence the integral converges.

10.3